

Student Number _____

ASCHAM SCHOOL

2014**YEAR 12****TRIAL****EXAMINATION**

Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks – 100

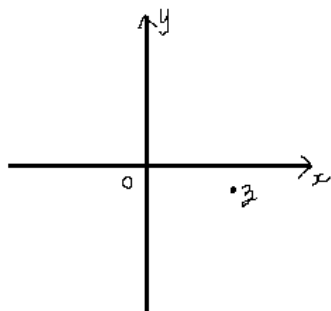
- Attempt Sections A and B.
- Section A is worth 10 marks.
- Answer Section A on the multiple choice answer sheet.
- Detach the multiple choice answer sheet from the back of the examination paper.
- Section B contains 6 questions worth 15 marks each.
- Answer each question in a new booklet.
- Label all sections clearly with your name/number and teacher.

SECTION A – 10 MULTIPLE CHOICE QUESTIONS

10 MARKS

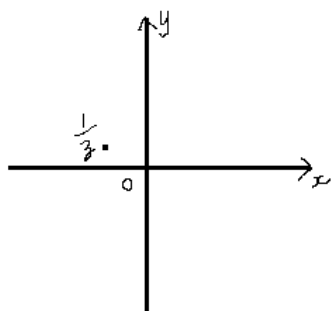
ANSWER ON THE ANSWER SHEET

- 1 The complex number z is sketched below.

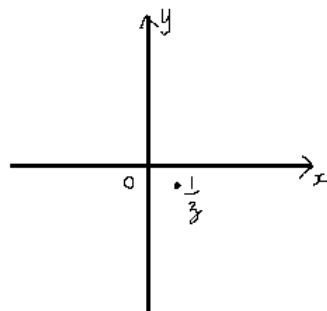


Which of the following sketches could describe $\frac{1}{z}$?

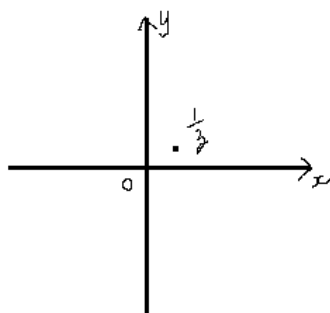
A



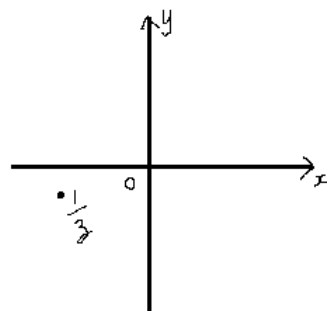
B



C



D



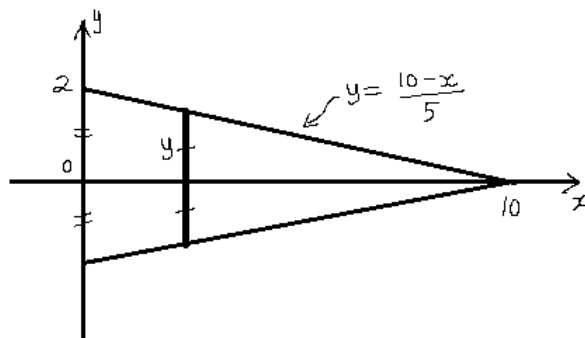
2 If z is complex, a solution to $z^5 = -1$ is:

- A $z = \text{cis} \frac{\pi}{5}$
 B $z = \text{cis} \frac{2\pi}{5}$
 C $z = \text{cis} \frac{8\pi}{5}$
 D $z = \text{cis} \frac{-2\pi}{5}$

3 If a particle of mass m is projected downwards under gravity and undergoes a resistive force of magnitude kv^3 then the acceleration is given by :

- A $\ddot{x} = g - kv^3$
 B $\ddot{x} = -g - kv^3$
 C $\ddot{x} = g - \frac{kv^3}{m}$
 D $\ddot{x} = -g - \frac{kv^3}{m}$

4 The volume of a kookaburra's beak is modelled by taking cross sections which are rhombuses, the longer diagonal of which is the length parallel to the y -axis between the lines shown. The smaller diagonal is half the length of the longer diagonal.



The volume would be given by:

- A $\int_0^{10} \frac{100 - 20x + x^2}{5} dx$
 B $\int_0^{10} \frac{100 - 20x + x^2}{10} dx$
 C $\int_0^{10} 4 - \frac{4x}{5} + \frac{x^2}{25} dx$
 D $\int_0^{10} 1 - \frac{x}{5} + \frac{x^2}{100} dx$

- 5 Let $f(x) = x^3 + x$ be an increasing function. Let $h(x)$ be the inverse function of $f(x)$. The point $(1, 2)$ lies on $y = f(x)$. The value of $h'(2) =$

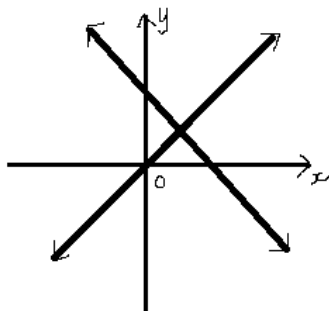
- A $\frac{1}{4}$
 B $\frac{1}{13}$
 C 4
 D 13

- 6 Consider the conic $\frac{x^2}{\lambda-9} + \frac{y^2}{\lambda-4} = 1$, where λ is a constant. If it is always an ellipse then:

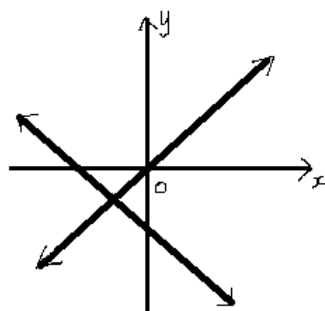
- A $\lambda < 4$
 B $4 < \lambda < 9$
 C $\lambda > 9$
 D $\lambda \leq 4$ or $\lambda \geq 9$.

- 7 The graph of $|x-1| = |y-1|$ could be:

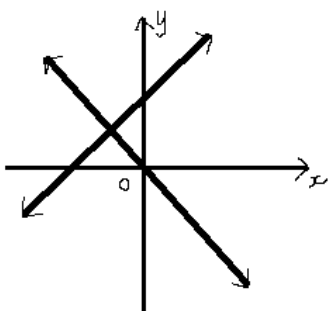
A



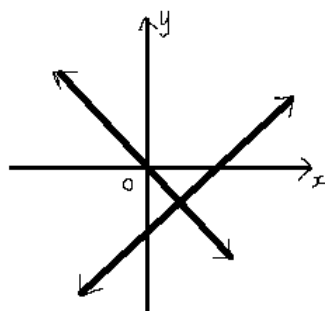
B



C



D



8 The solution to the inequality $\frac{1}{x} + \frac{1}{1-x} > 0$ is:

- A** $x < 0$
- B** $x > 1$
- C** $0 < x < 1$
- D** $x < 0$ or $x > 1$.

9 A possible factor of $15x^7 + 10x^5 - 2x^3 + 14$ would be:

- A** $2x - 7$
- B** $3x - 7$
- C** $5x + 3$
- D** $3x - 5$.

10 The value of $\lim_{x \rightarrow \infty} e^{-\frac{1}{x^2}} =$

- A** ∞
- B** 1
- C** -1
- D** 0.

SECTION 2 – 6 QUESTIONS EACH WORTH 15 MARKS

Question 11 – Begin a new writing booklet

a Find $\int \cot x \operatorname{cosec}^2 x \, dx$. 2

b Evaluate $\int_0^{\frac{1}{\sqrt{2}}} \frac{x}{\sqrt{1-x^4}} \, dx$. 3

c Use the substitution $u = \sqrt{e^x + 1}$ to find $\int \frac{e^{2x} \, dx}{\sqrt{e^x + 1}}$. 3

d

(i) Find constants A and B such that $\frac{\cos x}{1 - \sin^2 x} = \frac{A \cos x}{1 + \sin x} + \frac{B \cos x}{1 - \sin x}$. 2

(ii) Hence or otherwise find $\int \sec x \, dx$. 2

e Use the t – results to find $\int_0^{\frac{\pi}{3}} \frac{d\theta}{1 + \cos \theta}$. 3

Question 12 – Begin a new writing booklet

a Find a square root of $15 + 8i$. **2**

b Let $\alpha = 1 - i$ and $\beta = \sqrt{3} + i$.

i Find $\alpha\beta$ in Cartesian form. **1**

ii Express $\beta = \sqrt{3} + i$ in mod-arg form. **2**

iii If $\alpha = \sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right)$, find $\alpha\beta$ in mod-arg form. **1**

iv Hence find the exact value of $\cos \left(-\frac{\pi}{12} \right)$. **1**

c **2**

Sketch the locus of z if $\arg(z + 2) = \arg(z - 2i)$.

d Consider the quadrilateral $ABCD$ representing the complex numbers α, β, γ and δ , respectively.

Given that $\beta = \frac{1}{2}i\alpha$, $\alpha = -\gamma$, $|\beta| = |\delta|$ and $\arg \left(\frac{\beta}{\delta} \right) = \pi$,

(i) sketch the information on an Argand diagram, **1**

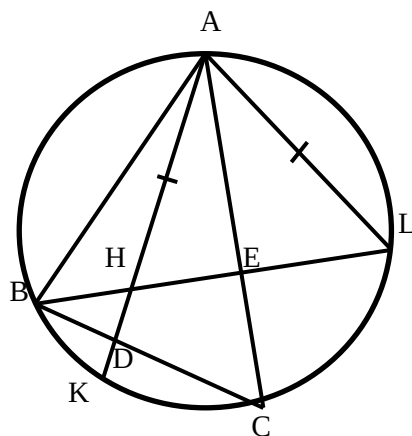
(ii) determine which type of quadrilateral is $ABCD$, giving reasons. **2**

e i Sketch $y = e^{2x} + 1$. **1**

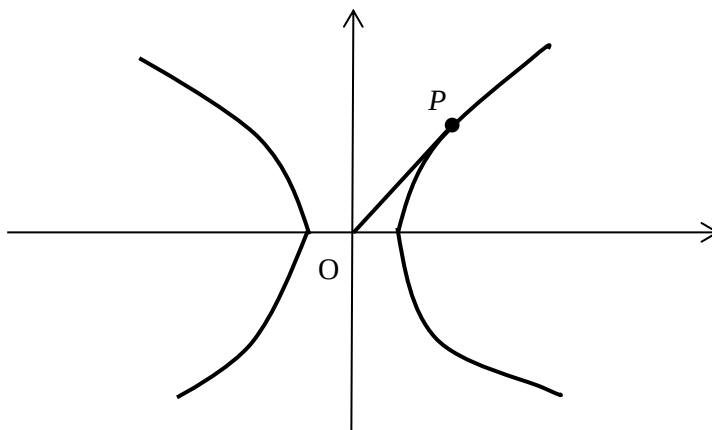
ii Hence sketch $y = \frac{e^{2x} + 1}{x}$. **2**

Question 13 – Begin a new writing booklet

- a** Given that $ALCK$ is a cyclic quadrilateral and H is a point on AK such that $AH = AL$. LH produced meets the circle again at B and meets AC at E . BC meets AK at D .



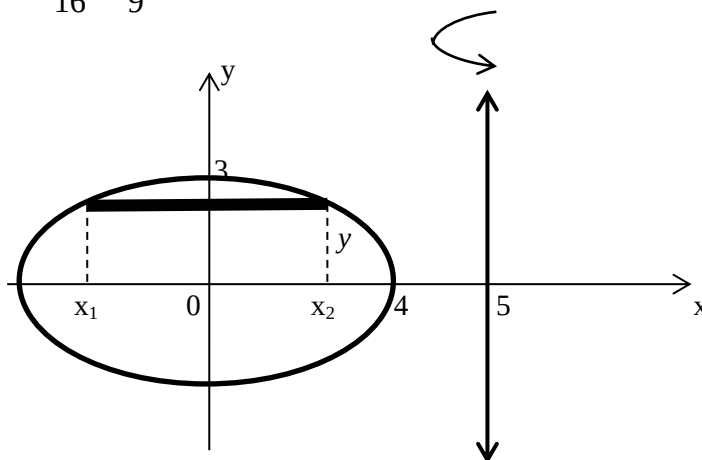
- i** Prove that $\angle AHL = \angle ACB$. 2
- ii** Hence state why that $HECD$ is a cyclic quadrilateral. 1
- iii** Given arc $KC = \text{arc } CL$, prove that HC is a diameter of $HECD$. 3
- b i** Show $\sqrt{(e^x + e^{-x})^2 - 4} = e^x - e^{-x}$ (Assume $x > 0$.) 2
- Consider the rectangular hyperbola $x^2 - y^2 = 1$ and the line segment OP .



- ii** Show that the point $P\left(\frac{e^x + e^{-x}}{2}, \frac{e^x - e^{-x}}{2}\right)$ lies on the hyperbola. 1
- iii** Using integration by parts (and then the table of standard integrals), show that 3
- $$\int \sqrt{x^2 - 1} \, dx = \frac{x\sqrt{x^2 - 1}}{2} - \frac{\ln(x + \sqrt{x^2 - 1})}{2} + C$$
- iv** Show that the area bounded by the hyperbola, OP and the x -axis is $\frac{x}{2}$ units². 3

Question 14 – Begin a new writing booklet
a

The ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is rotated about the line $x = 5$.



- (i) Show that the area of one annular type slice taken perpendicular to the axis of rotation is given by $A \approx \frac{80\pi}{3} \sqrt{9-y^2}$. 2
- (ii) Hence, by summing slices or otherwise, show that the volume of the resulting solid is $120\pi^2$ cubic units. 1

b

A particle of mass M is projected vertically upwards from O with initial speed I m/s. The particle is subjected to a constant gravitational force g m/s² downwards and a resistance of Mkv^2 , $k > 0$, where v is the speed at time t . Let x m be the displacement above O at time t seconds.

- i Show that the greatest height reached, H metres is given by $H = \frac{1}{2k} \ln \left(\frac{g + kI^2}{g} \right)$. 3
- ii The particle then begins to fall. Write down an equation for the acceleration $\ddot{x}$ on its downward journey and find the maximum speed the particle reaches on the downward journey, giving reasons. 2
- iii The particle returns to its point of projection with speed V m/s. Derive the equation for the distance travelled downwards and hence show that $(g + kI^2)(g - kV^2) = g^2$. 3

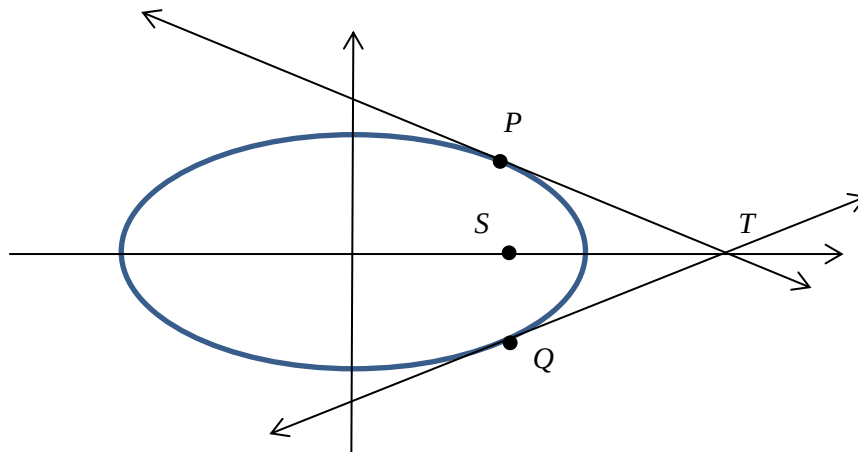
c

We know that $A^2 > 0, B^2 > 0, (A \pm B)^2 > 0$ for $A, B \neq 0$. Prove for $x, y \neq 0$:

- i $4x^2 + 6xy + 4y^2 > 0$, 2
- ii $3x^2 + 5xy + 3y^2 > 0$. 2

Question 15 – Begin a new writing booklet**a**

Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Tangents to the ellipse are drawn at $P(x_1, y_1)$ and $Q(x_2, y_2)$.



i Derive the equation of the tangent at P and state a similar result for the tangent at Q . 2

ii State the equation of the chord of contact from an external point $T(x_0, y_0)$. 1

The tangents meet at the point $T\left(\frac{a}{e}, 0\right)$.

iii Prove that the chord PQ passes through the focus S . 1

b

It is given that $\cos 5\theta = 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta$.

i Hence solve the equation $16x^5 - 20x^3 + 5x - 1 = 0$. 2

ii Show that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ and $\cos^2 \frac{2\pi}{5} \cos^2 \frac{4\pi}{5} = \frac{1}{16}$. 2

iii Hence deduce that the exact values are $\cos \frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4}$ and $\cos \frac{4\pi}{5} = \frac{-1-\sqrt{5}}{4}$. 2

Question 15 continues on the next page.

Question 15 continued

c

i **2**

Prove the identity $\frac{\sin\left(n + \frac{1}{2}\right)x}{2\sin\frac{x}{2}} + \cos(n+1)x = \frac{\sin\left(n + \frac{3}{2}\right)x}{2\sin\frac{x}{2}}.$

[Hint: $\cos(n+1)x = \cos\left(\left(n + \frac{1}{2}\right)x + \frac{1}{2}x\right)$]

ii **3**

Hence prove by Mathematical Induction (if $\sin\frac{x}{2} \neq 0$) that for $n = 1, 2, 3, \dots$

$$\frac{1}{2} + \cos x + \cos 2x + \cos 3x + \dots + \cos nx = \frac{\sin\left(n + \frac{1}{2}\right)x}{2\sin\frac{x}{2}}.$$

Question 16 – Begin a new writing booklet

- a** Solve $\cos 5x = \sin 9x$. 2
- b** The black-winged petrel from Lord Howe Island produces chicks of mass 30 grams. The approximate growth pattern of the chick then follows the equation
 $M = 4t + 30 + 8\sin \frac{\pi t}{2}$, where M is the mass of the chick in grams after t days,
 $0 \leq t \leq 70$. After 70 days the parents stop feeding the chick and it must then fend for itself.
- i** Between which two lines does this function lie? 2
- ii** What is the approximate maximum mass the chick can reach in this range? Give reasons. 2
- iii** Sketch the function and determine approximately how often the chicks get food. 2
- c** Consider the function $G_n = \int_0^\infty e^{-t} t^{n-1} dt$, where $n = 1, 2, 3, \dots$
- i** State an expression for G_{n+1} . 1
- ii** Use integration by parts to show that $G_{n+1} = nG_n$. 2
- iii** Show $G_1 = 1$. 2
- iv** Show $G_n = (n-1)!$ for all $n = 1, 2, 3, \dots$ 2

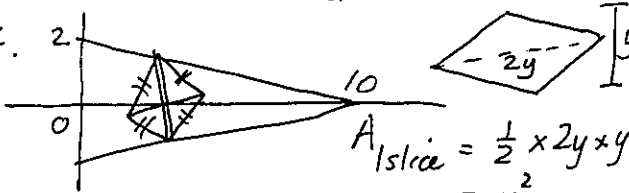
The end! 😊

MC: 1. $z = r \operatorname{cis} \theta \therefore \frac{1}{z} = \frac{1}{r} \operatorname{cis}(-\theta)$

reflect angle $\therefore$ (C)

2. $z^5 = -1 \therefore z_1 = \operatorname{cis} \frac{\pi}{5}$ then equally spaced, $\frac{2\pi}{5}$
 $= \operatorname{cis} \pi \therefore$ (A)

3. $\downarrow g$ $R = ma = -kr^3$
 $\therefore \ddot{x} = g - \frac{kr^3}{m} \therefore$ (C)

4. 
 $A_{\text{slice}} = \frac{1}{2} \times 2y \times y$
 $= y^2$
 $y^2 = \left(\frac{10-x}{5}\right)^2$
 $= \frac{100 - 20x + x^2}{25}$
 $V = \int_0^{10} y^2 dx$
 $\therefore$ (C)

5. $f(x) = x^3 + x \therefore$ let $y = x^3 + x$

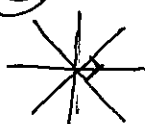
$\therefore$ Inverse of $y = x^3 + x$ is $\frac{dy}{dx} = 3x^2 + 1$
 $\therefore \frac{dx}{dy} = \frac{1}{3x^2 + 1}$
 $x = y^3 + y$

$\therefore y = h(x)$ so $h'(2)$ means when $x = 2$ is inverse in $x = y^3 + y \therefore y = 1$

So in original D & R swap so

$x = 1 \therefore \frac{dx}{dy} = \frac{1}{3(1)^2 + 1} = \frac{1}{4} \therefore$ (A)

6. $\frac{x^2}{\lambda - 9} + \frac{y^2}{\lambda - 4} = 1$ ellipse then $\lambda - 9 > 0$ and $\lambda - 4 > 0$
 $\therefore \lambda > 9$ and $\lambda > 4$
 $\therefore \lambda > 9 \therefore$ (C)

7. $|x-1| = |y-1|$ $|x| = |y|$ 
 Move centre to (1,1). $\therefore$ (A)

8. $\frac{1}{x} + \frac{1}{1-x} > 0, x \neq 0, x \neq 1$
 Test $\frac{-}{0} \frac{+}{1} \frac{-}{\infty}$
 $\frac{1-x+x}{x(1-x)} > 0 \therefore 0 < x < 1$
 $\frac{1}{x(1-x)} > 0 \therefore$ (C)

MC cont'd:

9. Factor $ax-b$, "a" factor of $15x^7$, b factor of 14 $\therefore 3x-7 \therefore$ (B)

10. $\lim_{x \rightarrow \infty} e^{-\frac{1}{x^2}} \frac{1}{x^2} \rightarrow 0$ as $x \rightarrow \infty$
 $\therefore e^{-0} = e^0 = 1 \therefore$ (B)

Q11 a) $\int \cot x \operatorname{cosec}^2 x dx$
 $= - \int \cot x \operatorname{cosec}^2 x dx$ (2)
 $= - \frac{\cot^2 x}{2} + C$

b) $\int_0^{\frac{1}{\sqrt{2}}} \frac{x}{\sqrt{1-x^4}} dx = \frac{1}{2} \int_0^{\frac{1}{\sqrt{2}}} \frac{2x}{\sqrt{1-(x^2)^2}} dx$
 $= \frac{1}{2} \left[\sin^{-1}(x^2) \right]_0^{\frac{1}{\sqrt{2}}}$
 $= \frac{1}{2} \left[\sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}(0) \right]$
 $= \frac{1}{2} \left[\frac{\pi}{6} - 0 \right]$
 $= \frac{\pi}{12}$ (3)

c) $\int \frac{e^{2x} dx}{\sqrt{e^x + 1}} = \int \frac{e^x \cdot e^x dx}{\sqrt{e^x + 1}}$ let $u = \sqrt{e^x + 1}$
 $du = \frac{1}{2}(e^x + 1)^{-\frac{1}{2}} \cdot e^x dx$
 $= \int (u^2 - 1) \cdot 2 du$
 $= \frac{2u^3}{3} - 2u + C$ (3)
 $= \frac{2(\sqrt{e^x + 1})^3}{3} - 2\sqrt{e^x + 1} + C$
 $u^2 = e^x + 1$
 $e^x = u^2 - 1$
 $2du = \frac{e^x dx}{\sqrt{e^x + 1}}$

d) i) Find $A \Delta B$: (2)

$\frac{\cos x}{1 - \sin^2 x} = \frac{A \cos x}{1 + \sin x} + \frac{B \cos x}{1 - \sin x}$
 $\frac{\cos x}{1 - \sin^2 x} = \frac{A(1 - \sin x) \cos x + B(1 + \sin x) \cos x}{(1 + \sin x)(1 - \sin x)}$
 $\therefore \cos x \equiv \cos x (A - A \sin x + B + B \sin x)$
 $\therefore A + B = 1, B - A = 0 \therefore A = B = \frac{1}{2}$

Q11 cont'd:

$$d) i) \therefore \frac{\cos x}{1-\sin^2 x} = \frac{1}{2} \left(\frac{\cos x}{1+\sin x} + \frac{\cos x}{1-\sin x} \right)$$

$$\therefore ii) \int \sec x dx = \int \frac{\cos x}{\cos^2 x} dx$$

$$= \int \frac{\cos x}{1-\sin^2 x} dx \quad (2)$$

$$= \frac{1}{2} \left[\ln(1+\sin x) - \ln(1-\sin x) \right] + C$$

$$= \ln \sqrt{\frac{1+\sin x}{1-\sin x}} + C$$

$$e) \text{ let } t = \tan \frac{\theta}{2}$$

$$dt = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$$

$$\therefore \int_0^{\frac{\pi}{3}} \frac{d\theta}{1+\cos \theta}$$

$$\therefore 2dt = (t^2+1)d\theta$$

$$\therefore d\theta = \frac{2dt}{t^2+1}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{t^2+1}$$

$$\theta = \frac{\pi}{3} \quad t = \frac{1}{\sqrt{3}}$$

$$\theta = 0 \quad t = 0$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{1+\frac{1-t^2}{1+t^2}}$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2dt}{2} \quad (3)$$

$$= \int_0^{\frac{1}{\sqrt{3}}} 1 dt$$

$$= \left[t \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= \frac{1}{\sqrt{3}}$$

Q12

$$a) \text{ If } \sqrt{15+8i} = a+ib \text{ where } a, b \in \mathbb{R}$$

$$\text{then } 15+8i = (a+ib)^2 = a^2-b^2+2abi$$

$$\text{Equating real \& imaginary parts:}$$

$$a^2-b^2=15, \quad 8=2ab \Rightarrow 4=ab \quad (2)$$

$$\text{By inspection, } a=4, b=1 \text{ or } a=-4, b=-1$$

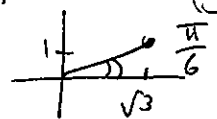
$$\therefore \text{A root is } 4+i \text{ (or } -4-i).$$

Q12 cont'd

$$b) \alpha = 1-i, \beta = \sqrt{3}+i$$

$$i) \alpha\beta = (1-i)(\sqrt{3}+i) = \sqrt{3}+1+i(1-\sqrt{3}) \quad (1)$$

$$ii) \beta = \sqrt{3}+i = 2 \cos \frac{\pi}{6} \quad |\beta| = \sqrt{3^2+1^2} = 2 \quad (2)$$



$$iii) \alpha = \sqrt{2} \cos \left(-\frac{\pi}{4} \right)$$

$$\alpha\beta = 2 \cos \frac{\pi}{6} \times \sqrt{2} \cos \left(-\frac{\pi}{4} \right) \quad (1)$$

$$= 2\sqrt{2} \cos \left(\frac{\pi}{6} - \frac{\pi}{4} \right)$$

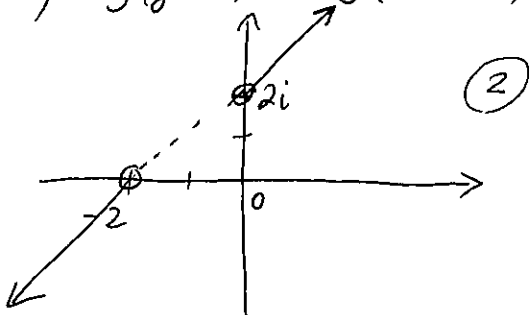
$$= 2\sqrt{2} \cos \left(\frac{-\pi}{12} \right) = 2\sqrt{2} \left(\cos \left(\frac{-\pi}{12} \right) + i \sin \left(\frac{-\pi}{12} \right) \right)$$

$$iv) = \sqrt{3}+1+i(1-\sqrt{3}) \text{ Equating reals:}$$

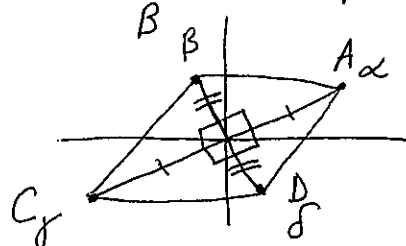
$$2\sqrt{2} \cos \left(\frac{-\pi}{12} \right) = \sqrt{3}+1 \quad (1)$$

$$\therefore \cos \left(\frac{-\pi}{12} \right) = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$c) \arg(z+2) = \arg(z-2i)$$



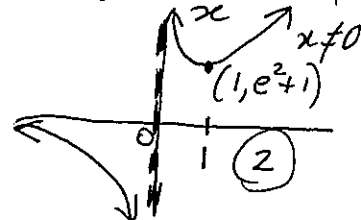
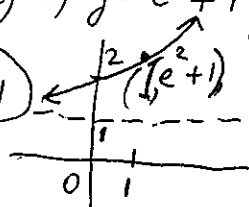
$$d) \beta = \frac{1}{2} i \alpha, \alpha = -\delta, |\beta| = |\delta|, \arg \left(\frac{\beta}{\delta} \right) = \pi$$



ABCD is a rhombus since diagonals bisect at $\frac{1}{2}$ and that's it! (2)

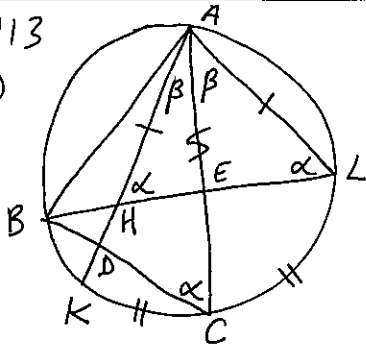
$$e) i) y = e^{2x} + 1 \quad (1)$$

$$y = e^{2x} + 1$$



Q13

a)

Let $\angle AHL = \alpha$.i) $\angle ALH = \alpha$ (base $\angle$ s of isosceles $\triangle AHL$ are equal)Also $\angle ACB = \alpha$ ($\angle$ s standing at

(2) Circumference on arc AB equal)

 $\therefore \angle AHL = \angle ACB$ (both equal $\angle ALH$)ii) $\therefore HECD$ is a cyclic quad(1) ($\angle AHL = \angle ACB$, exterior $\angle$ of quad $HECD =$ interior opposite)iii) In $\triangle AHE$ and $\triangle ALE$,1. AE common2. $\angle HAE = \angle LAE$ (equal arcs subtend equal $\angle$ s at circumference)

(3)

3. $AH = AL$ (given) $\therefore \triangle AHE \equiv \triangle ALE$ (SAS) $\therefore \angle AEH = \angle AEL$ (matching $\angle$ s in congruent $\triangle$ s)but $\angle AEH + \angle AEL = 180^\circ$ (straight line) $\therefore 2x = 180$ ($x = \angle AEH$) $x = 90$ $\therefore \angle AEH = 90^\circ$ then $\angle HEC = 90^\circ$ (vertically opposite $\angle$ s) $\therefore HC$ is diameter ($\angle$ in semicircle is 90°)b) i) $\sqrt{(e^x + e^{-x})^2 - 4} = \sqrt{e^{2x} + 2 + e^{-2x} - 4}$

(2)

$$= \sqrt{e^{2x} - 2 + e^{-2x}}$$

$$= \sqrt{(e^x - e^{-x})^2}$$

$$= e^x - e^{-x} \text{ if } x > 0$$

ii) Sub: $\left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$ (1)

$$= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4}$$

$$= \frac{4}{4} = 1, \therefore \text{lies on } x^2 - y^2 = 1.$$

Q13 cont'd: $\int \sqrt{x^2 - 1} dx$ let $u = \sqrt{x^2 - 1}$

$$\text{iii) } uv - \int v du$$

$$du = \frac{1}{2}(x^2 - 1)^{-\frac{1}{2}} \cdot 2x dx$$

$$= x\sqrt{x^2 - 1} - \int \frac{x \cdot x}{\sqrt{x^2 - 1}} dx$$

$$dv = 1 dx$$

$$v = x$$

$$= x\sqrt{x^2 - 1} - \int \frac{x^2}{\sqrt{x^2 - 1}} dx \quad (3)$$

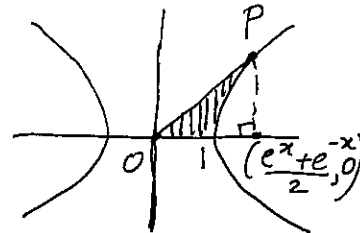
$$= x\sqrt{x^2 - 1} - \int \frac{x^2 - 1}{\sqrt{x^2 - 1}} + \frac{1}{\sqrt{x^2 - 1}} dx$$

$$= x\sqrt{x^2 - 1} - \int \sqrt{x^2 - 1} dx + \int \frac{1}{\sqrt{x^2 - 1}} dx$$

$$\therefore 2 \int \sqrt{x^2 - 1} dx = x\sqrt{x^2 - 1} - \ln(x + \sqrt{x^2 - 1}) + C$$

$$\therefore \int \sqrt{x^2 - 1} dx = \frac{x\sqrt{x^2 - 1}}{2} - \frac{\ln(x + \sqrt{x^2 - 1})}{2} + C$$

iv)



$$x^2 - y^2 = 1$$

$$y^2 = x^2 - 1$$

$$y = \pm \sqrt{x^2 - 1}$$

$$\frac{e^x + e^{-x}}{2}$$

$$\text{Area} = \text{Triangle} - \int_1^{\frac{e^x + e^{-x}}{2}} \sqrt{x^2 - 1} dx$$

$$= \frac{1}{2}bh - \int_1^{\frac{e^x + e^{-x}}{2}} \sqrt{x^2 - 1} dx$$

$$= \frac{1}{2} \left(\frac{e^x + e^{-x}}{2} \right) \left(\frac{e^x - e^{-x}}{2} \right) -$$

$$\left[\frac{x\sqrt{x^2 - 1}}{2} - \frac{\ln(x + \sqrt{x^2 - 1})}{2} \right]_1^{\frac{e^x + e^{-x}}{2}}$$

$$= \frac{1}{2} \left(\frac{e^{2x} - e^{-2x}}{4} \right) \left(\frac{e^x + e^{-x}}{4} \sqrt{\left(\frac{e^x + e^{-x}}{2} \right)^2 - 1} \right.$$

$$\left. - \ln \left(\frac{e^x + e^{-x}}{2} + \sqrt{\left(\frac{e^x + e^{-x}}{2} \right)^2 - 1} \right) \right]$$

$$- \left[\frac{\sqrt{1^2 - 1}}{2} - \frac{\ln(1 + \sqrt{1^2 - 1})}{2} \right] \quad (3)$$

$$= \frac{e^{2x} - e^{-2x}}{8} - \left(\frac{e^x + e^{-x}}{4} \sqrt{\frac{e^{2x} + 2 + e^{-2x}}{4} - 4} \right.$$

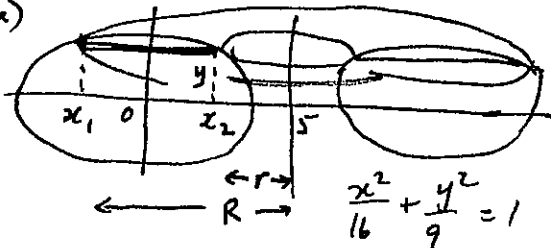
$$\text{Using (i) } - \ln \left(\frac{e^x + e^{-x}}{2} + \sqrt{\frac{e^{2x} + 2 + e^{-2x}}{4} - 4} \right) - 0$$

$$= \frac{e^{2x} - e^{-2x}}{8} - \left(\frac{e^x + e^{-x}}{4} \times \frac{e^x - e^{-x}}{2} - \ln \left(\frac{e^x + e^{-x}}{2} + \frac{e^x - e^{-x}}{2} \right) \right)$$

$$= \frac{e^{2x} - e^{-2x}}{8} - \frac{e^{2x} - e^{-2x}}{8} + \frac{\ln 2}{2}$$

$$= 0 + \frac{\ln e^x}{2} = \frac{x \ln e}{2} = \frac{x}{2} \text{ unit}^2$$

Q14 a)



i) Area of 1 slice $\div \pi R^2 - \pi r^2$ (2)
 $= \pi (R+r)(R-r)$

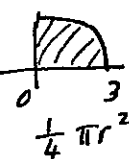
$= \pi ((5-x_1) + (5-x_2))((5-x_1) - (5-x_2))$
 $= \pi (10 - (x_1+x_2))(x_2-x_1)$

Now make x subject to find x_1, x_2 :

$\frac{x^2}{16} = 1 - \frac{y^2}{9} \Rightarrow x = \pm \sqrt{16(1 - \frac{y^2}{9})}$
 $= \pm \frac{4}{3} \sqrt{9-y^2}$

so $x_1 = -\frac{4}{3} \sqrt{9-y^2}$ $x_2 = +\frac{4}{3} \sqrt{9-y^2}$

$\therefore A \div \pi (10 - (0)) (2 \times \frac{4}{3} \sqrt{9-y^2})$
 $\div \pi \times \frac{80}{3} \sqrt{9-y^2}$ #

ii) $V \div \int_{y=-3}^3 \frac{80\pi}{3} \sqrt{9-y^2} dy$ 
 $= 2 \int_0^3 \frac{80\pi}{3} \sqrt{9-y^2} dy$
 $= 2 \times \frac{80\pi}{3} \times \frac{1}{4} \pi 3^2$
 $= 120\pi^2 \text{ unit}^3$ (1)

OR Using Pappas' Theorem:

$V = \text{Area of Cross-section} \times \text{Average Radius} \times \text{Circumference}$
 $= \pi \times 3 \times 4 \times 2\pi \times 5$
 $= 120\pi^2 \text{ u}^3$

b) i) $F = ma = -Mg - Mkv^2$
 $v = \int -g - Mkv^2$
 $\therefore \ddot{x} = -g - kv^2$
 $v \frac{dv}{dx} = \frac{-g - kv^2}{v}$
 $\therefore \int \frac{2k dx dv}{dv} = \int \frac{-2kv}{g + kv^2} dv$
 $\therefore 2kx = -[\ln(g + kv^2)]$ (3)

Q14 b) i) cont'd $x = \frac{-1}{2k} [\ln(g + kv^2) - \ln(g + kI^2)]$
 $\therefore H = \frac{1}{2k} \ln \left(\frac{g + kI^2}{g} \right)$

ii) $F = ma = Mg - Mkv^2$
 $\therefore \ddot{x} = g - kv^2$ (2)
 Max speed should be terminal speed
 as starting from rest $\therefore \ddot{x} = 0 = g - kv^2$
 $\therefore v = \sqrt{\frac{g}{k}}$ since $v > 0$ (speed)

iii) $v = V$ when $x = H$ down.

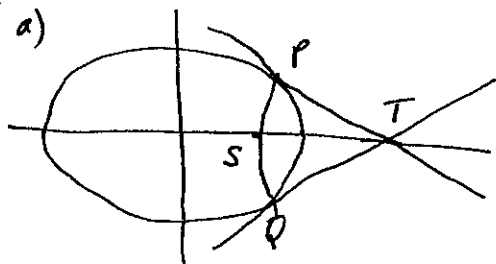
$\therefore \ddot{x} = g - kv^2$
 $v \frac{dv}{dx} = g - kv^2$
 $\frac{dv}{dx} = \frac{g - kv^2}{v}$
 $\therefore \int_0^V \frac{-2k dx}{dv} = \int_0^V \frac{-2kv}{g - kv^2} dv$
 $x = \frac{-1}{2k} [\ln(g - kv^2)]_0^V$
 $= \frac{-1}{2k} [\ln(g - kV^2) - \ln(g - k0^2)]$
 $H = \frac{1}{2k} \ln \left(\frac{g}{g - kV^2} \right)$

Distance up = distance down
 $\therefore \frac{1}{2k} \ln \left(\frac{g + kI^2}{g} \right) = \frac{1}{2k} \ln \left(\frac{g}{g - kV^2} \right)$
 $(g + kI^2)(g - kV^2) = g^2$ #

c) i) $4x^2 + 6xy + 4y^2 = 3x^2 + 3y^2 + 6xy + x^2 + y^2$
 $= 3(x^2 + 2xy + y^2) + x^2 + y^2$ (2)
 $= 3(x+y)^2 + x^2 + y^2$

> 0 if $x, y \neq 0$ since $(x+y)^2 > 0$
 and $x^2, y^2 > 0$
 ii) $3x^2 + 5xy + 3y^2 = 2x^2 + 4xy + 2y^2 + x^2 + xy + y^2$
 $= 2(x+y)^2 + \frac{x^2}{2} + \frac{y^2}{2} + \frac{x^2 + 2xy + y^2}{2}$
 $= 2(x+y)^2 + \frac{x^2}{2} + \frac{y^2}{2} + \frac{(x+y)^2}{2}$ (2)
 > 0 since all terms > 0 for $x, y \neq 0$
 $\emptyset \in \mathbb{D}$.

Q15 a)



$$i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{b^2 x}{a^2 y} \text{ at } x=x_1, m = -\frac{b^2 x_1}{a^2 y_1}$$

$\therefore$ Eqn of tangent is:

$$y - y_1 = m(x - x_1)$$

$$y - y_1 = -\frac{b^2 x_1}{a^2 y_1} (x - x_1) \quad (2)$$

$$\frac{y y_1}{b^2} - \frac{y_1^2}{b^2} = -\frac{x x_1}{a^2} + \frac{x_1^2}{a^2}$$

$$\therefore \frac{x x_1}{a^2} + \frac{y y_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$$

since (x_1, y_1) lies on ellipse

$$\therefore \text{Eqn of tangent is } \frac{x x_1}{a^2} + \frac{y y_1}{b^2} = 1 \text{ at } P$$

Similarly tangent at Q is

$$\frac{x x_2}{a^2} + \frac{y y_2}{b^2} = 1.$$

ii) Eqn of chord of contact is

$$\frac{x x_0}{a^2} + \frac{y y_0}{b^2} = 1. \quad (1)$$

iii) chord of contact from $T(\frac{a}{e}, 0)$ is chord PQ - sub $(\frac{a}{e}, 0) \Rightarrow$

$$\frac{x(\frac{a}{e})}{a^2} + \frac{y(0)}{b^2} = 1 \quad (1)$$

$$\therefore \frac{x}{ae} = 1 \Rightarrow x = ae \text{ i.e. PQ}$$

is line $x = ae \therefore S(ae, 0)$ lies on vertical PQ. QED.

$$b) \cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta.$$

$$i) 16x^5 - 20x^3 + 5x - 1 = 0 \text{ if } x = \cos\theta$$

then equivalent to $\cos 5\theta = 1$

$$\therefore 5\theta = 0 + 2k\pi \text{ where } k \in \mathbb{Z}$$

$$\theta = \frac{2k\pi}{5} \text{ i.e. } \theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \dots$$

etc. so $x = \cos(\frac{2k\pi}{5})$ where values are 5 (distinct) roots of equation

$$\therefore \cos 0, \cos \frac{2\pi}{5}, \cos(\frac{4\pi}{5}), \cos(\frac{6\pi}{5}), \cos(\frac{8\pi}{5})$$

$$\cos \frac{6\pi}{5}, \cos \frac{8\pi}{5}, \cos \frac{10\pi}{5}, \dots$$

$$\text{(Distinct) roots are } \cos 0, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos \frac{6\pi}{5}, \cos \frac{8\pi}{5} \quad (2)$$

$$\text{but (ii) } \cos(-\frac{4\pi}{5}) = \cos(\frac{4\pi}{5}), \cos \frac{2\pi}{5} = \cos(-\frac{2\pi}{5})$$

$$\therefore \text{Sum of roots} = -\frac{b}{a} = \frac{0}{16} = 0$$

$$\therefore \cos 0 + \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} + \cos(-\frac{2\pi}{5}) + \cos(-\frac{4\pi}{5}) = 0$$

$$\therefore 1 + 2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5} = 0$$

$$\therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

$$\text{and product} = -\frac{c}{a} = \frac{1}{16} \quad (2)$$

$$\therefore 1 \times \cos \frac{2\pi}{5} \times \cos \frac{4\pi}{5} \times \cos(-\frac{2\pi}{5}) \times \cos(-\frac{4\pi}{5}) = \frac{1}{16}$$

$$\therefore \cos^2 \frac{2\pi}{5} \cos^2 \frac{4\pi}{5} = \frac{1}{16}$$

$$iii) \text{ let } \cos \frac{2\pi}{5} = a, \cos \frac{4\pi}{5} = b$$

$$\therefore a + b = -\frac{1}{2} \quad a^2 b^2 = \frac{1}{16} \quad (2)$$

$$ab = \pm \frac{1}{4}$$

$$\text{but } \cos \frac{4\pi}{5} < 0 \text{ so } ab = -\frac{1}{4} \text{ only}$$

$$\therefore a = -\frac{1}{4b} \therefore -\frac{1}{4b} + b = -\frac{1}{2}$$

$$\therefore 1 - 4b^2 = 2b \Rightarrow 4b^2 + 2b - 1 = 0$$

$$\Rightarrow b = \frac{-2 \pm \sqrt{20}}{8} = \frac{-1 \pm \sqrt{5}}{4} \text{ but } b < 0 \therefore b = \frac{-1 - \sqrt{5}}{4}$$

Q15 cont'd:

b iii) cont'd $\therefore a = -\frac{1}{2} - \left(\frac{-1-\sqrt{5}}{4}\right)$
 $= \frac{-1+\sqrt{5}}{4}$
 $\therefore \cos \frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4}, \quad \cos \frac{4\pi}{5} = \frac{-1-\sqrt{5}}{4}$

c) i) RTP: $\frac{\sin(n+\frac{1}{2})x}{2\sin \frac{x}{2}} + \cos(n+1)x = \frac{\sin(n+\frac{3}{2})x}{2\sin \frac{x}{2}}$

Proof: $\frac{\sin(n+\frac{1}{2})x}{2\sin \frac{x}{2}} + \cos(n+1)x$
 $= \frac{\sin(n+\frac{1}{2})x}{2\sin \frac{x}{2}} + \cos((n+\frac{1}{2})x + \frac{1}{2}x)$ (2)

$= \frac{\sin(n+\frac{1}{2})x}{2\sin \frac{x}{2}} + \cos(n+\frac{1}{2})x \cos \frac{1}{2}x - \sin(n+\frac{1}{2})x \sin \frac{1}{2}x$

$= \frac{\sin(n+\frac{1}{2})x}{2\sin \frac{x}{2}} + 2\sin \frac{x}{2} \cos(n+\frac{1}{2})x \cos \frac{1}{2}x - 2\sin(n+\frac{1}{2})x \sin \frac{1}{2}x$

$= \frac{\sin(n+\frac{1}{2})x + \cos(n+\frac{1}{2})x \sin x - 2\sin(n+\frac{1}{2})x \sin \frac{1}{2}x}{2\sin \frac{x}{2}}$

$= \frac{\sin(n+\frac{1}{2})x [1 - 2\sin^2 \frac{1}{2}x] + \cos(n+\frac{1}{2})x \sin x}{2\sin \frac{x}{2}}$

$= \frac{\sin(n+\frac{1}{2})x \cos x + \cos(n+\frac{1}{2})x \sin x}{2\sin \frac{x}{2}}$

$= \frac{\sin(n+\frac{1}{2}+1)x}{2\sin \frac{x}{2}} = \frac{\sin(n+\frac{3}{2})x}{2\sin \frac{x}{2}}$ (PE)

ii) RTP: let $P(n)$ be the proposition that $\frac{1}{2} + \cos x + \cos 2x + \cos 3x + \dots + \cos nx = \frac{\sin(n+\frac{1}{2})x}{2\sin \frac{x}{2}}$

Proof: let $P(1)$ be $\frac{1}{2} + \cos x = \frac{\sin(\frac{3}{2}x)}{2\sin \frac{x}{2}}$

LHS = $\frac{1}{2} + \cos x$
 $= \frac{1}{2} + \cos(\frac{1}{2}x + \frac{1}{2}x)$ [or use (i) result where $n=0$]

$= \frac{1}{2} + \cos^2 \frac{1}{2}x - \sin^2 \frac{1}{2}x$
 $= \frac{1 + 2\cos^2 \frac{1}{2}x - 2\sin^2 \frac{1}{2}x}{2} \times \frac{\sin x}{\sin \frac{x}{2}}$

$= \frac{\sin \frac{x}{2} + 2\cos^2 \frac{1}{2}x \sin \frac{x}{2} - 2\sin^2 \frac{1}{2}x \sin \frac{x}{2}}{2\sin \frac{x}{2}}$

$= \frac{\sin \frac{x}{2} (1 - 2\sin^2 \frac{1}{2}x) + 2\sin \frac{x}{2} \cos^2 \frac{1}{2}x}{2\sin \frac{x}{2}}$

$= \frac{\sin \frac{x}{2} \cos x + \sin x \cos \frac{x}{2}}{2\sin \frac{x}{2}} = \frac{\sin(\frac{x}{2} + x)}{2\sin \frac{x}{2}} = \text{RHS}$

 $\therefore P(1)$ true.

Assume $P(k)$ true i.e. $k \in \mathbb{N}, k \geq 1$
 $\frac{1}{2} + \cos x + \cos 2x + \dots + \cos kx = \frac{\sin(k+\frac{1}{2})x}{2\sin \frac{x}{2}}$

RTP: $P(k+1)$ true i.e.

$\frac{1}{2} + \cos x + \cos 2x + \dots + \cos kx + \cos(k+1)x$
 $= \frac{\sin(k+1+\frac{1}{2})x}{2\sin \frac{x}{2}}$ (3)

Proof: Consider the LHS of $P(k+1)$:

$\frac{1}{2} + \cos x + \cos 2x + \dots + \cos kx + \cos(k+1)x$
 $= \frac{\sin(k+\frac{1}{2})x}{2\sin \frac{x}{2}} + \cos(k+1)x$ using $P(k)$

$= \frac{\sin(k+\frac{3}{2})x}{2\sin \frac{x}{2}}$ from (i)

 $=$ RHS of $P(k+1) \therefore P(k+1)$ true. $\therefore P(n)$ is true for all $n \in \mathbb{N}, n \geq 1$ by Math Induction.

Q16

a) $\cos 5x = \frac{1}{2} \sin 9x$ (2)

$\therefore \cos 5x = \cos(\frac{\pi}{2} - 9x)$

$\therefore 5x = 2k\pi \pm (\frac{\pi}{2} - 9x) \quad k \in \mathbb{Z}$

$\therefore 5x = 2k\pi + \frac{\pi}{2} - 9x$ OR $5x = 2k\pi - (\frac{\pi}{2} - 9x)$

$\therefore 14x = 2k\pi + \frac{\pi}{2}$ OR $4x = -(2k\pi - \frac{\pi}{2})$

$\therefore x = \frac{2k\pi + \frac{\pi}{2}}{14}$ OR $x = -\frac{(4k\pi - \frac{\pi}{2})}{4}$

$\therefore x = \frac{\pi(4k+1)}{28}$ OR $x = -\frac{\pi(4k-1)}{4}$

b) $M = 4t + 30 + 8 \sin \frac{\pi t}{2}, 0 \leq t \leq 70$

i) $-1 \leq \sin \frac{\pi t}{2} \leq 1$

$\therefore -8 \leq 8 \sin \frac{\pi t}{2} \leq 8$

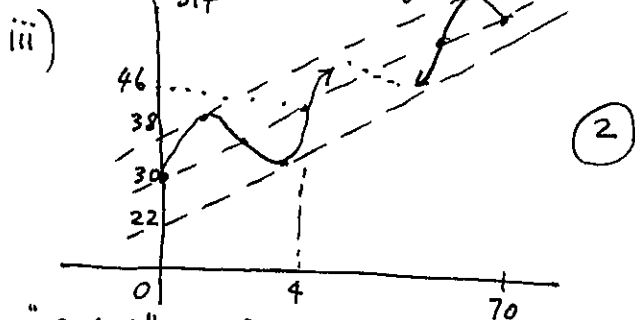
$\therefore 4t + 30 - 8 \leq 4t + 30 + 8 \sin \frac{\pi t}{2} \leq 4t + 30 + 8$

$\therefore 4t + 22 \leq y \leq 4t + 38$ (2)

$\therefore$ Lines are $y = 4t + 22$
 and $y = 4t + 38$

16 b) ii) If $t=70$ $y = 4(70) + 30 + 8\sin(35\pi)$
 $= 310 + 0 = 310$
 But if $t=69$ $y = 4(69) + 30 + 8\sin(\frac{68\pi}{2})$
 $= 306 + 8 \times 1$
 $= 314$

$\therefore$ Max mass is on Day 69 - 314 g.



"Period" is 4 days so chick probably gets fed about every 4 days.

c) $G_n = \int_0^\infty e^{-t} t^{n-1} dt \quad n=1,2,3,\dots$

i) $G_{n+1} = \int_0^\infty e^{-t} t^n dt$ (1)

ii) $G_n = \int_0^\infty e^{-t} t^{n-1} dt$
 $= uv - \int v du$ where $u = e^{-t}$ and $du = -e^{-t} dt$
 $= \left[e^{-t} \frac{1}{n} t^n \right]_0^\infty - \int_0^\infty \frac{1}{n} t^n \cdot (-e^{-t}) dt$ where $dv = t^{n-1} dt$ and $v = \frac{1}{n} t^n$
 $= 0 - \frac{1}{n} t^n e^{-t} \Big|_0^\infty + \frac{1}{n} \int_0^\infty t^n \cdot e^{-t} dt$
 $= 0 + \frac{1}{n} \int_0^\infty e^{-t} t^n dt$

$\therefore G_n = \frac{1}{n} G_{n+1}$ (2)

$\therefore G_{n+1} = n G_n$ QED

iii) $G_1 = \int_0^\infty e^{-t} t^{1-1} dt$
 $= \int_0^\infty e^{-t} \cdot 1 dt$
 $= \left[-e^{-t} \right]_0^\infty$ (2)

$= -0 - (-e^{-0})$
 $= e^0 = 1$ QED

16 cont'd

c) iii) RTP: $G_n = (n-1)!$

Now $G_1 = 1$ and $G_{n+1} = n G_n$.

$\therefore G_2 = 1 \cdot G_1 = 1 \times 1 = 1$

$G_3 = 2 \cdot G_2 = 2 \cdot 1$

$G_4 = 3 \cdot G_3 = 3 \cdot 2 \cdot 1 = 3!$

$\therefore G_5 = 4 \cdot G_4 = 4 \cdot 3 \cdot 2 \cdot 1 = 4!$

$\vdots$

$G_n = (n-1) G_{n-1}$

$= (n-1) \cdot (n-2) \cdot (n-3) \cdot \dots \cdot 2 \cdot 1$

$\therefore G_n = (n-1)! \quad n=1,2,3,\dots$ QED

The End! ☹️